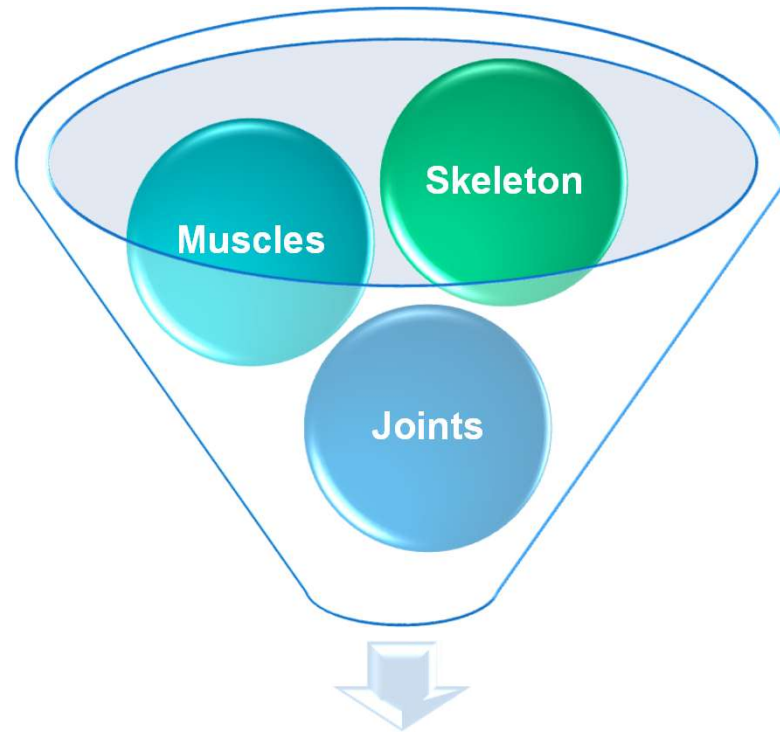


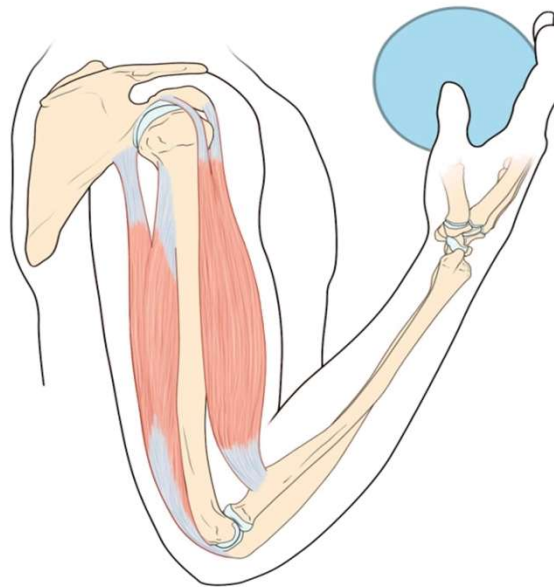


3.2 THE MUSCULOSKELETAL SYSTEM

- **Human Skeleton**
- **Types of Muscle**
- **Joints**



Movement

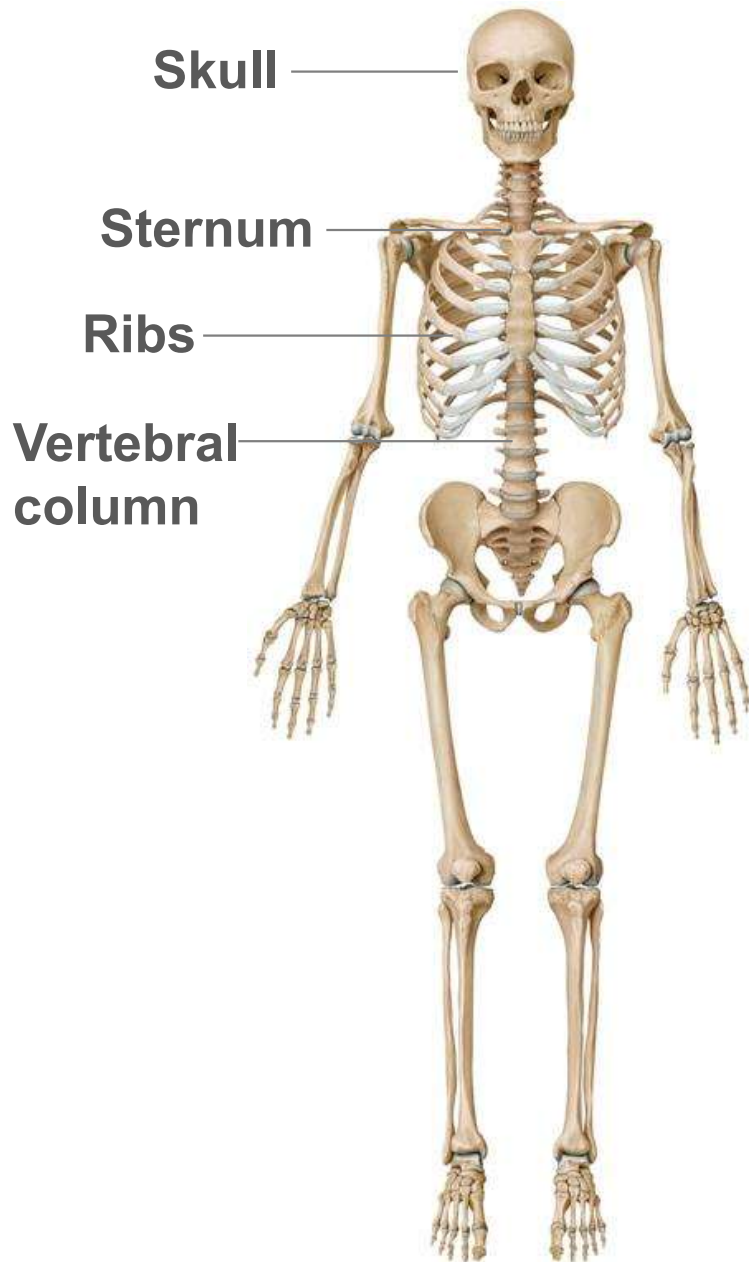


HUMAN SKELETON

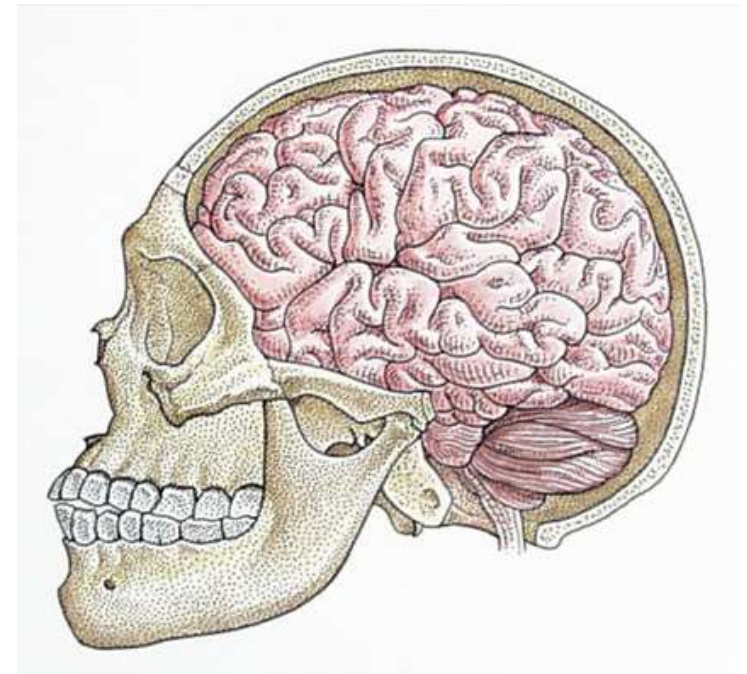
- Divisions
 - Axial
 - Appendicular
- Approx 206 bones
 - Shape
 - Classification
- Tissue composition



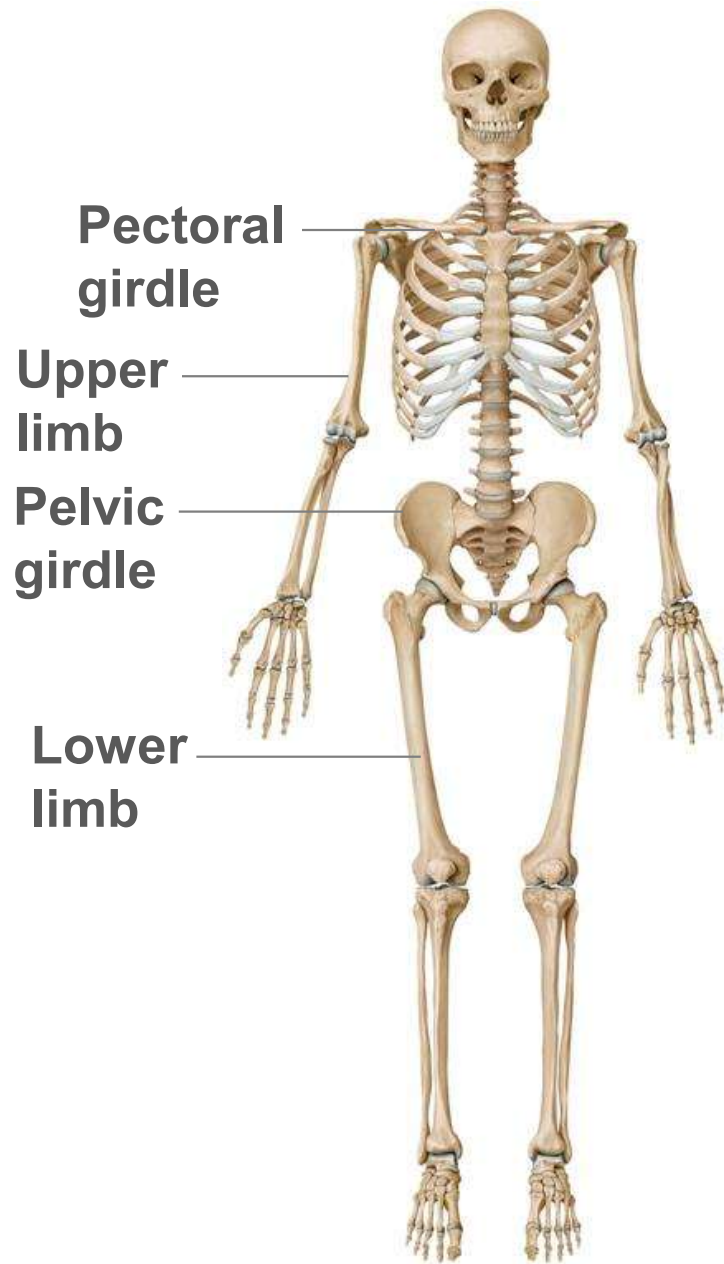
AXIAL SKELETON



- 80 bones
- Supports, stabilizes, and protects vital organs



APPENDICULAR SKELETON



- 126 bones
- Responsible for a large portion of movement



Skull

Sternum

Ribs

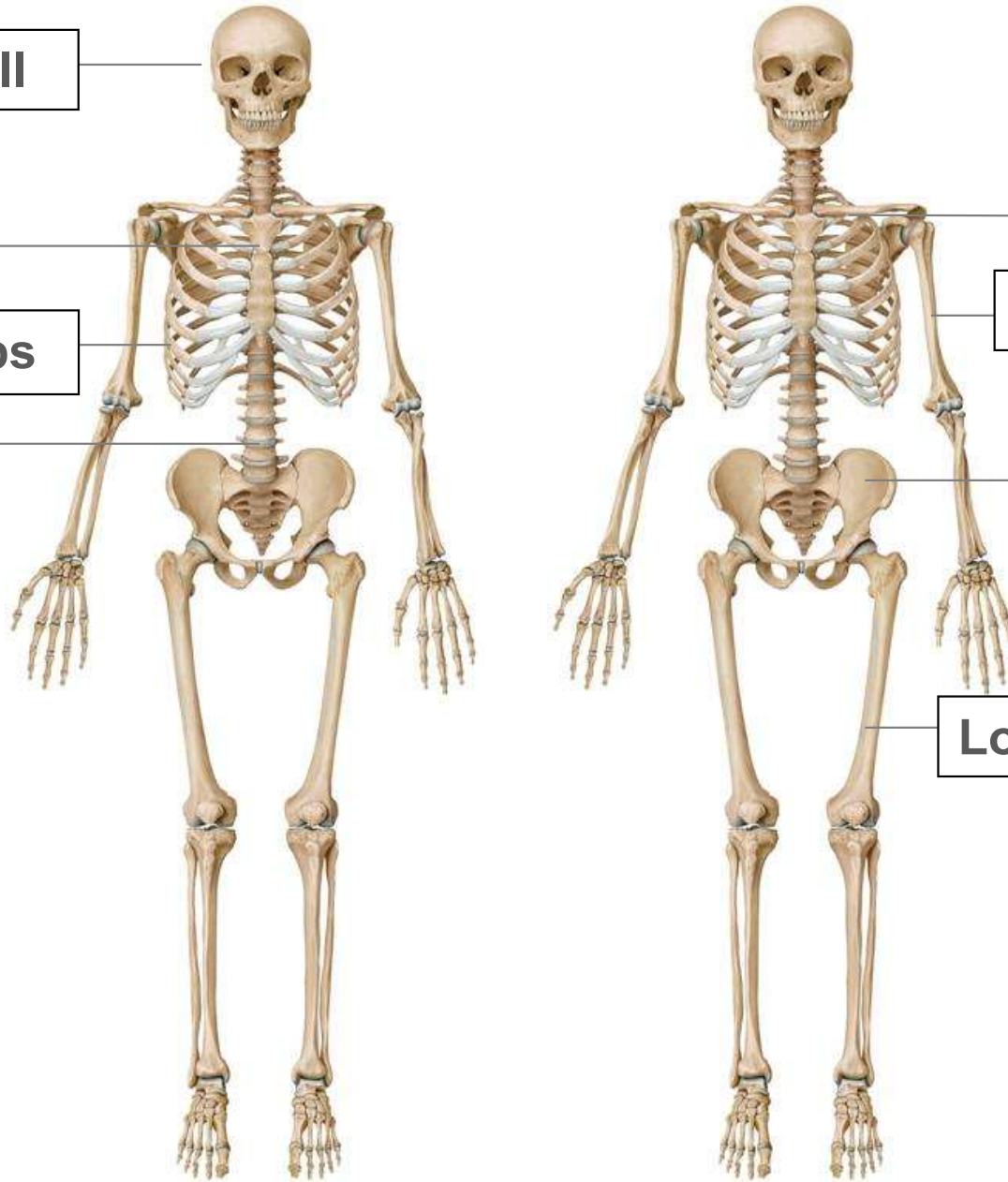
**Vertebral
column**

Pectoral girdle

Upper limb

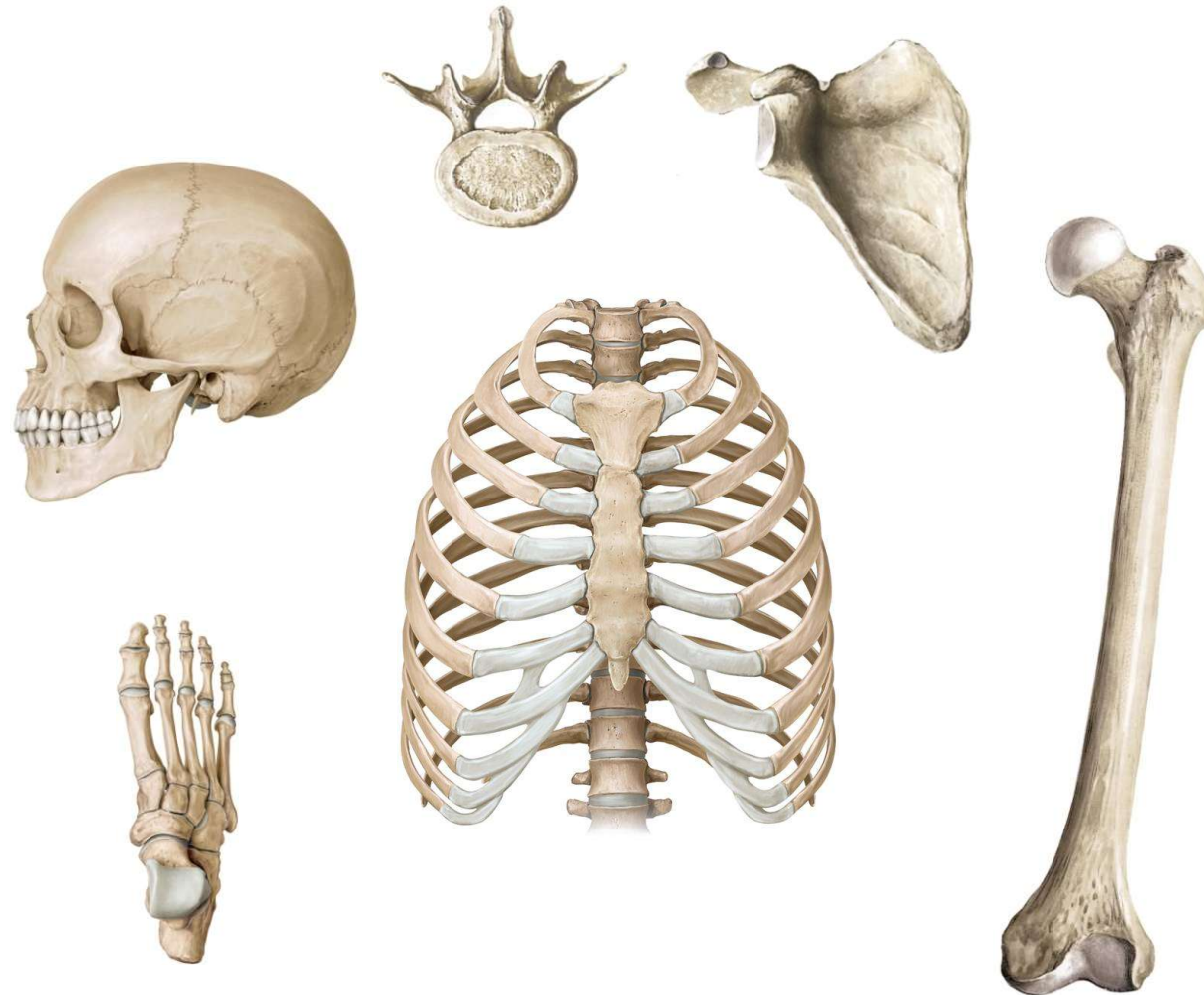
Pelvic girdle

Lower limb



BONE SHAPES

Shape determines function

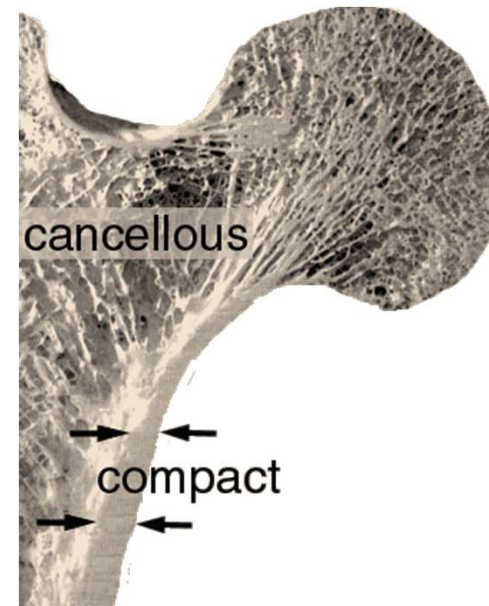
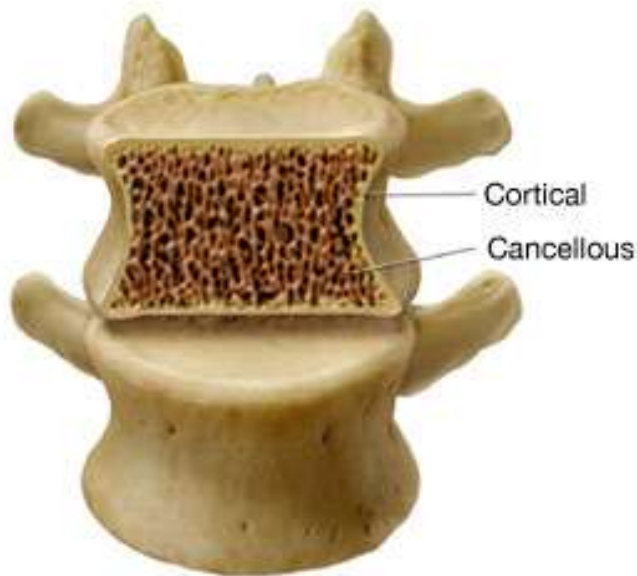


Shape	Examples		Function
Short	Carpals 	Tarsals 	Shock absorbers
Long	Femur 	Humerus 	Levers
Flat	Skull 	Scapula 	Protect organs
		Ribs 	
Irregular	Facial bones 	Vertebrae 	Special function
Sesamoid	Patella 		Change pressure / friction

BONE CLASSIFICATION

2 classes of bone tissue:

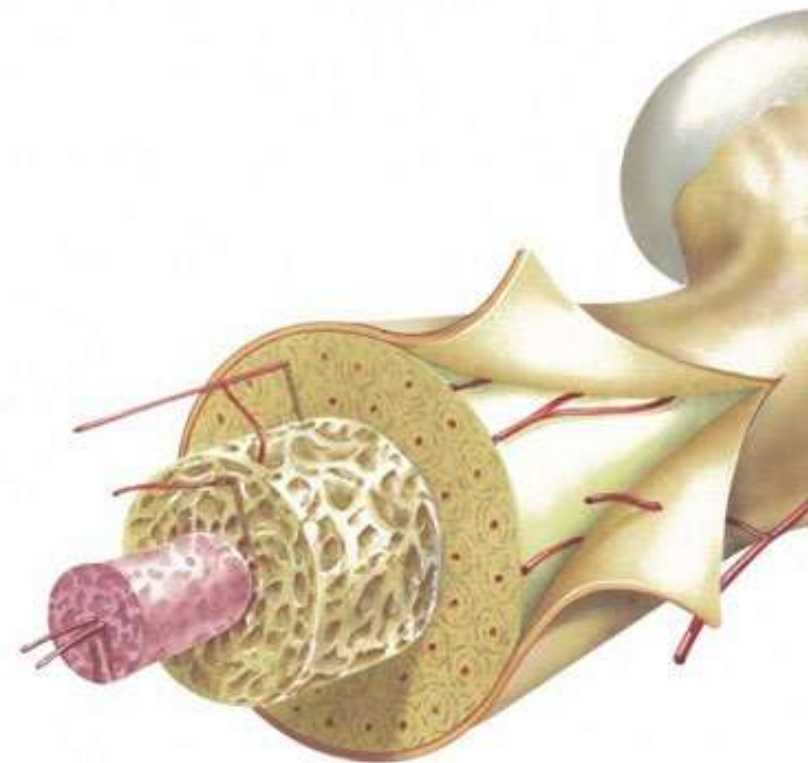
- Compact / cortical
- Spongy / cancellous



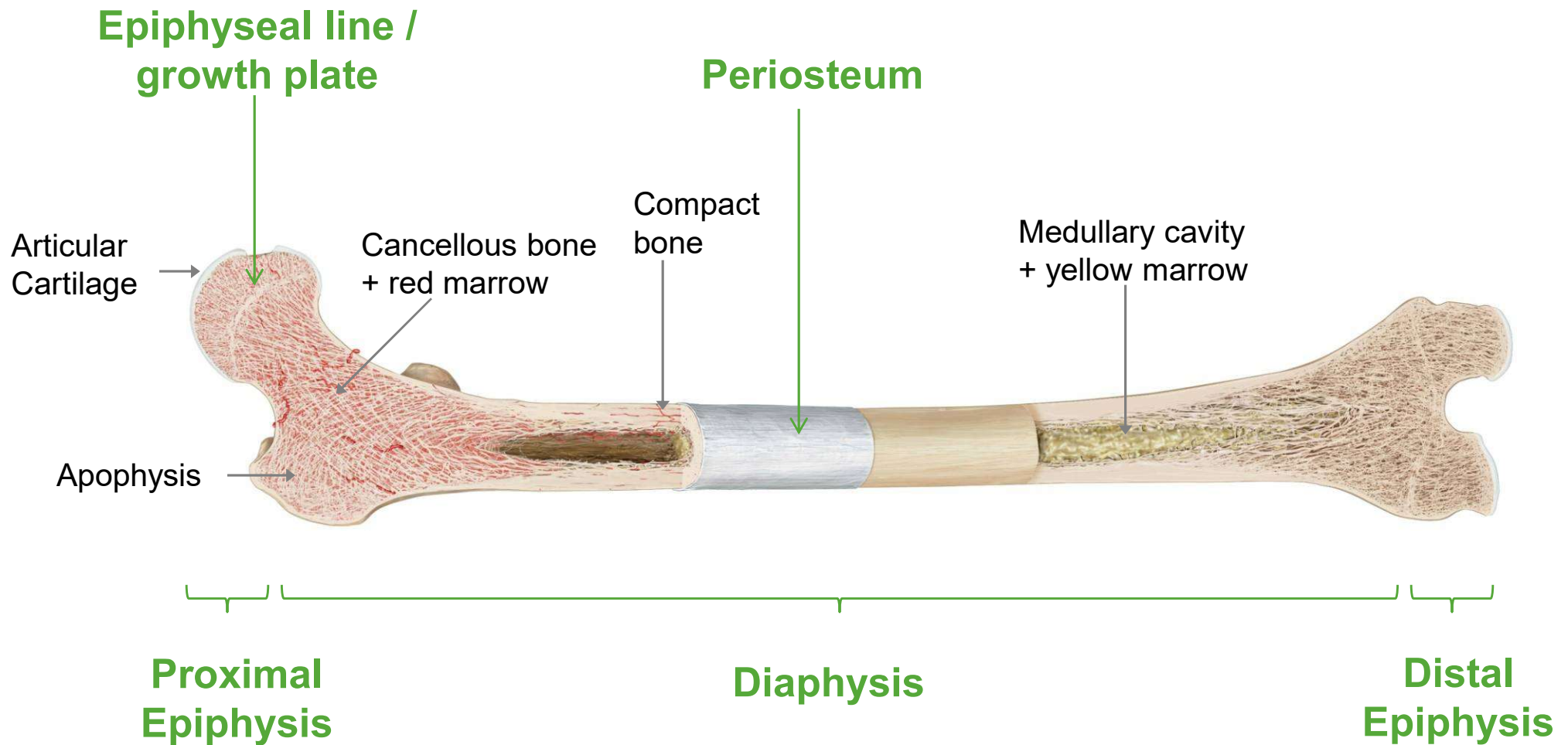
	Cancellous	Compact
Porosity	High: low mineral density and high collagen	Low: high mineral density and low collagen
Structure	Honey comb	Compact
		
Characteristic	Flexible but is not stress resistant	Stiff and stress resistant but less flexible
Function	Shock absorption	Withstanding stress
Location	Vertebrae	Long bones

BONE COMPOSITION

- **Calcium carbonate and calcium phosphate**
 - 60-70% of bone
 - Stiffness
 - Resistance to pressing forces
- **Collagen protein**
 - Flexibility
 - Resist pulling forces
 - When lost bone becomes brittle
- **Water**
 - 20% of bone (vs. 60% of body)



LONG BONES

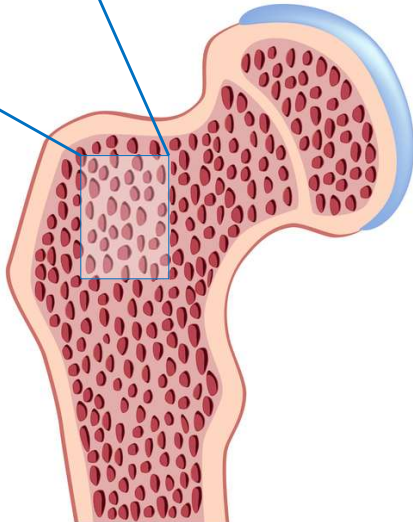


BONE AND EXERCISE

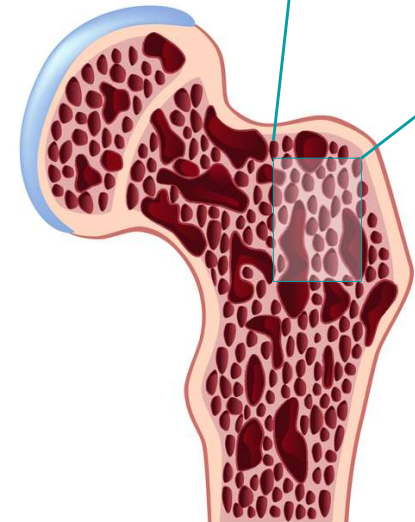
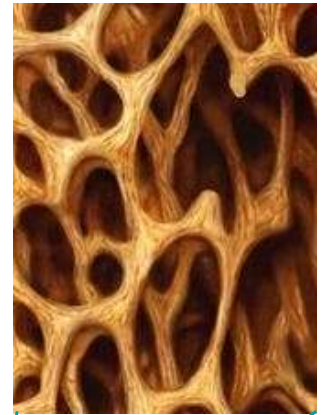
Strong bone at young age and
with regular exercise



Density
Mineralization
Porosity

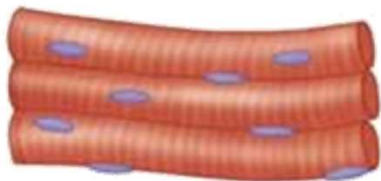
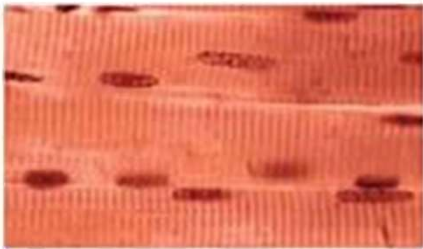


Weak bone with ageing and
without regular exercise

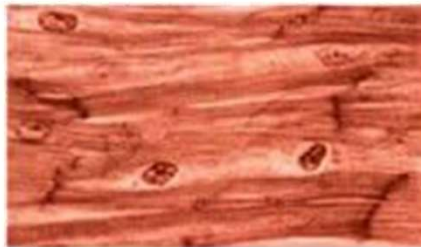
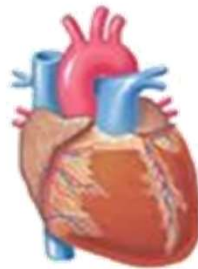


MUSCLE TYPES

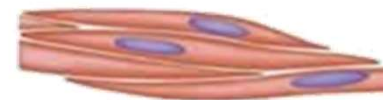
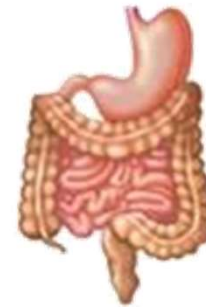
1. Skeletal

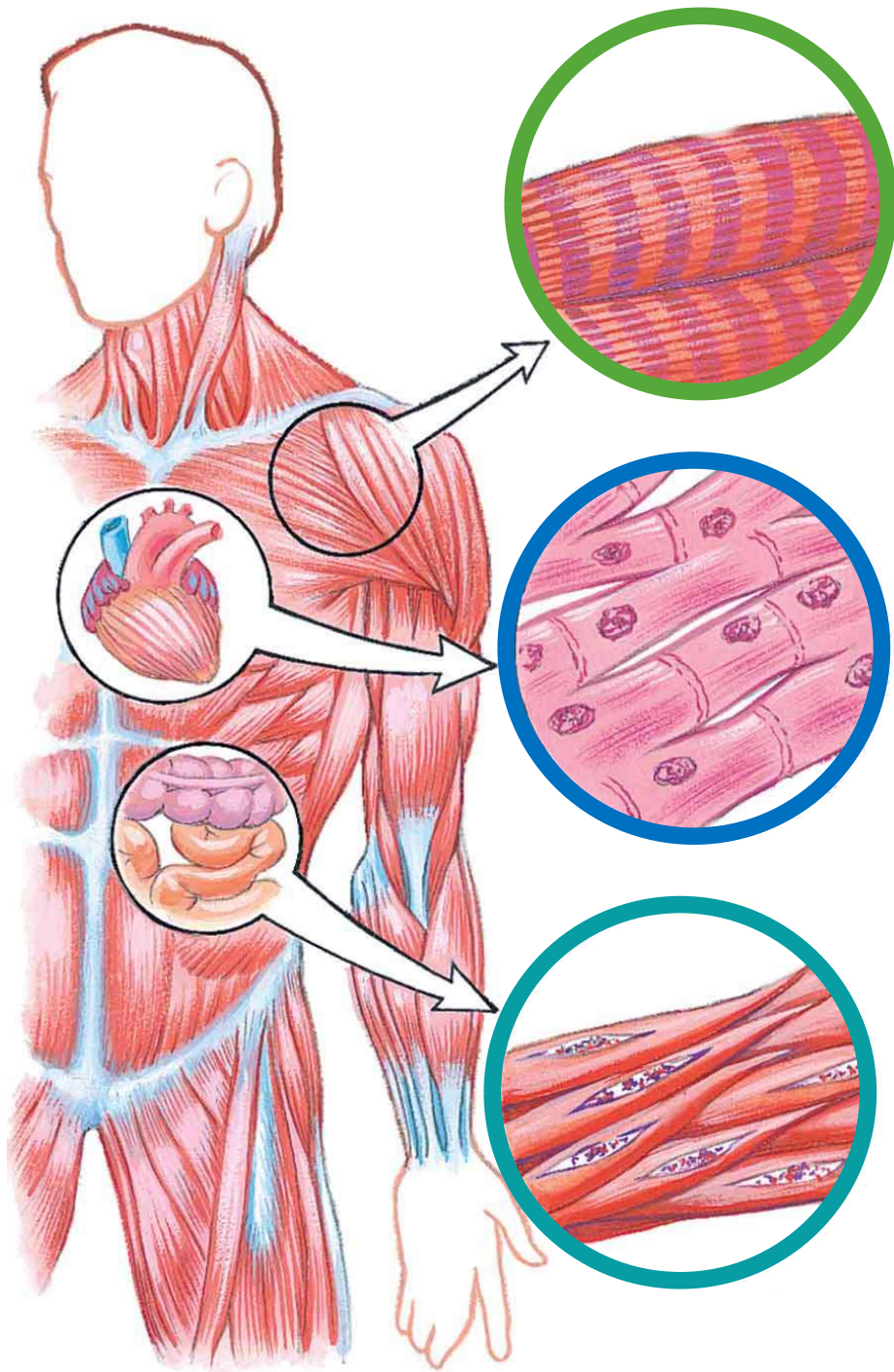


2. Cardiac



3. Smooth





CARDIAC MUSCLE

- Heart contraction / beating
- Very fatigue resistant
- Has own intrinsic beat
- Autonomic nerve control / involuntary

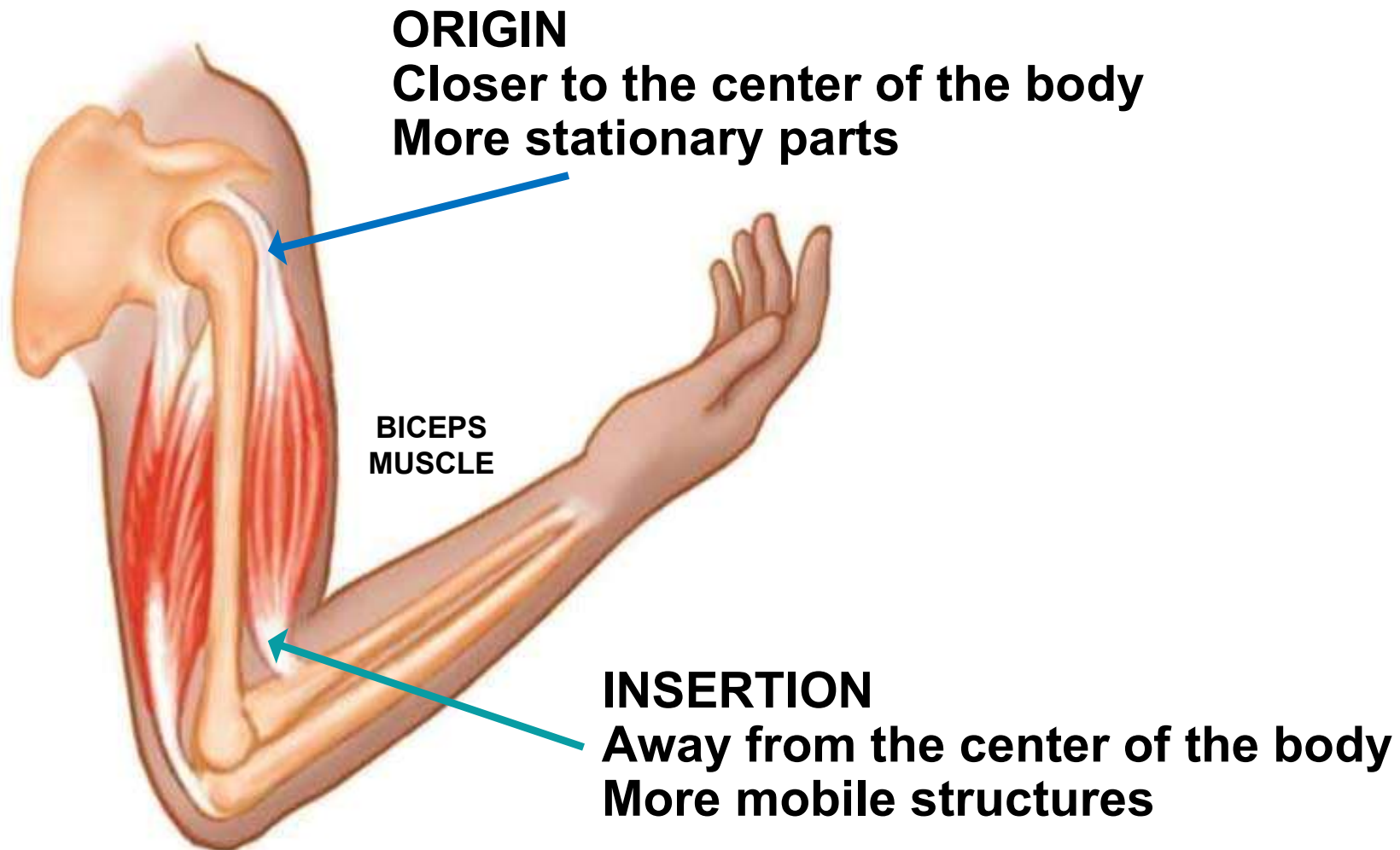
SKELETAL MUSCLE

- Attached to bone
- Contraction = body movement
- Motor nerve control / voluntary

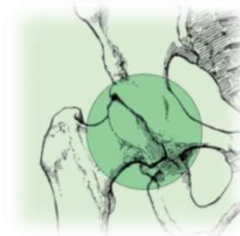
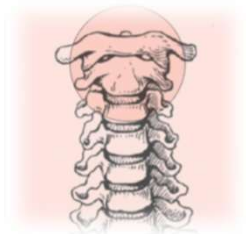
SMOOTH MUSCLE

- Blood vessels & organs
- Slow and uniform contractions
- Fatigue resistant
- Autonomic nerve control / involuntary

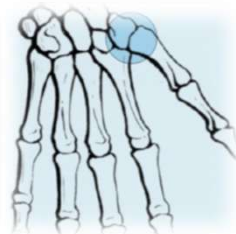
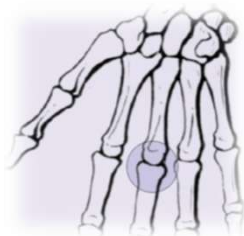
SKELETAL MUSCLE CHARACTERISTICS



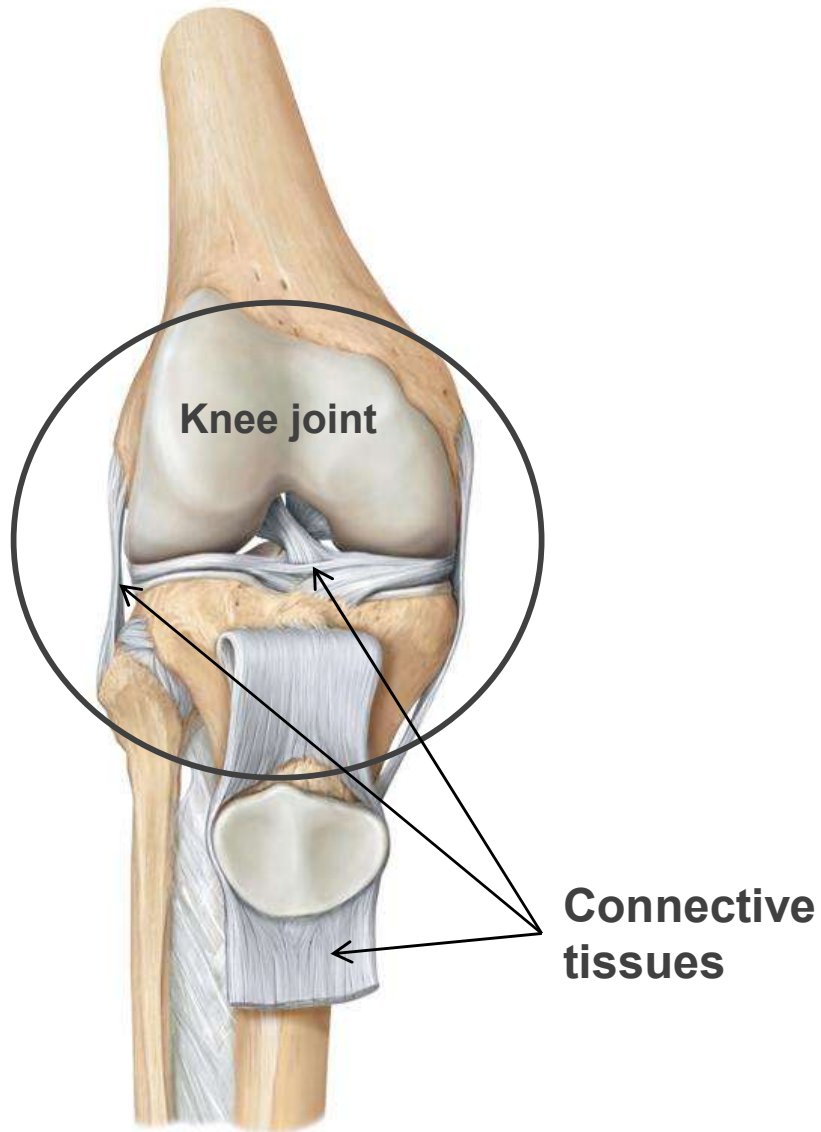
JOINTS



- **Joint classification**
- **Synovial joints**



WHAT'S A JOINT?



- Connection between two or more bones
- Strands of connective tissue ensure stability
- Classified by the degree of movement

Fibrous joints

- No movement
- Absorb shock
- Example: skull sutures



Cartilaginous joints

- Limited movement
- Absorb shock
- Example: intervertebral discs

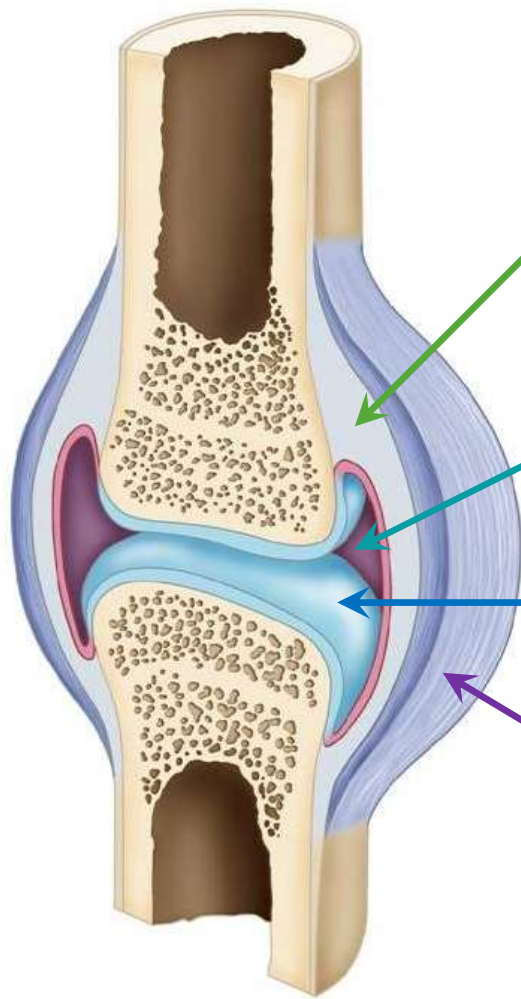


Synovial joints

- Greatest degree of movement
- Allow movement, most common
- Example: hip joint



SYNOVIAL JOINTS



Joint Capsule

- Surrounds the joint and provides support
- Lined with synovial membrane that secretes synovial fluid

Joint cavity

- Filled with synovial fluid for lubrication
- Also cushions

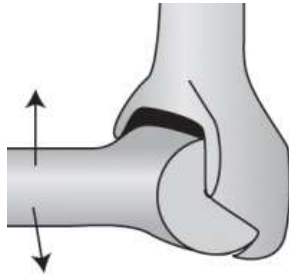
Hyaline cartilage

- Dense white connective tissue that covers and protects the ends of the articulating bones

Ligaments, extrinsic (and intrinsic)

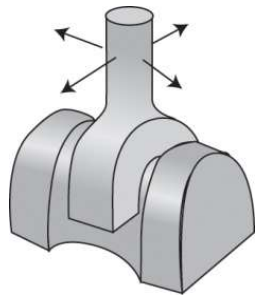
- Support the joint
- Connect the articulating bones of the joint

JOINT MOVEMENTS



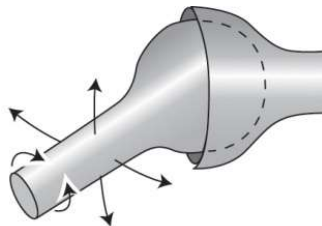
Uniaxial

- Movement about one axis



Biaxial joints

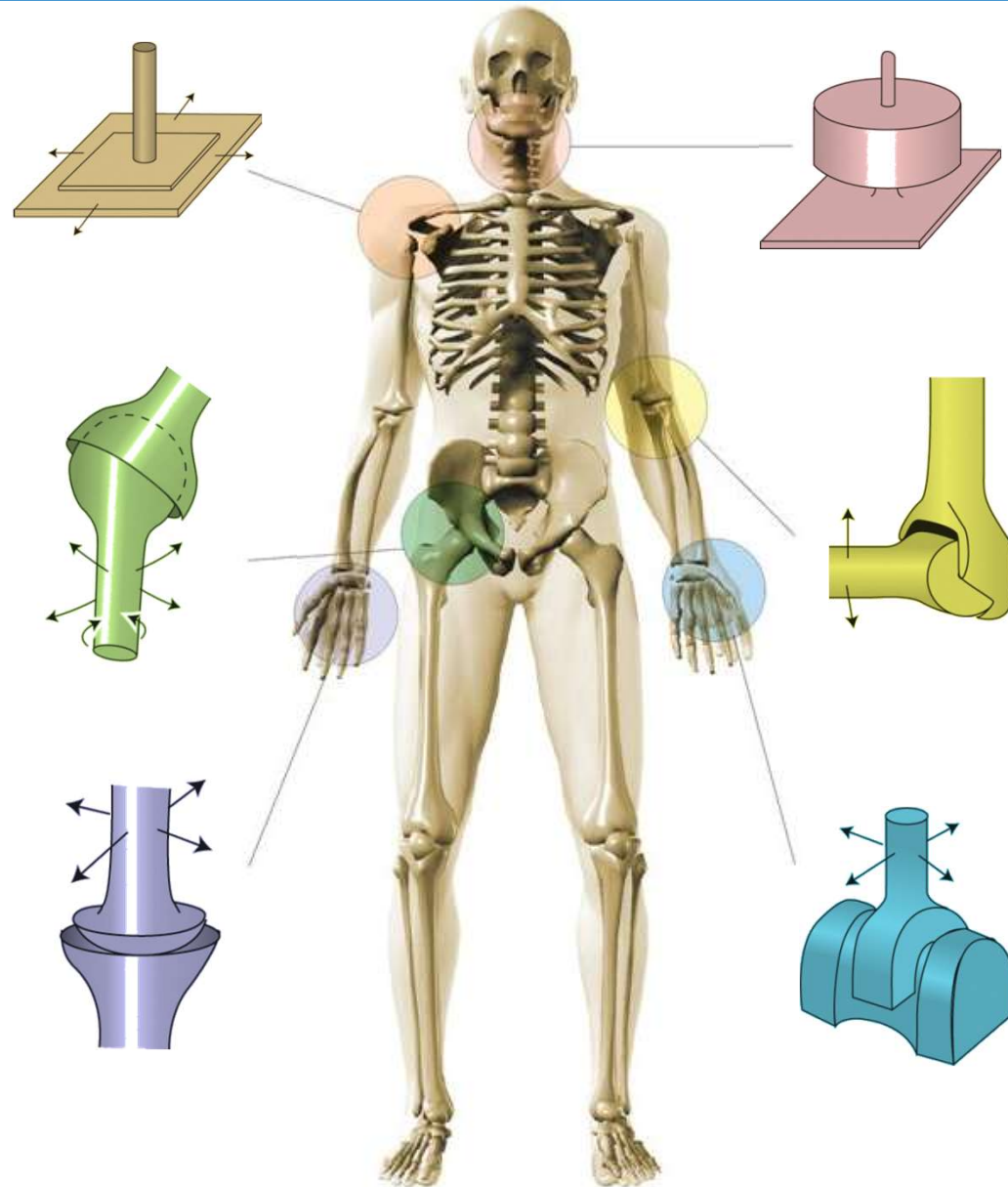
- Movement about two perpendicular axes

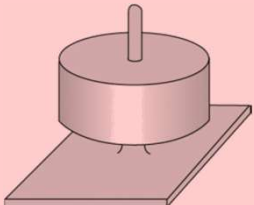
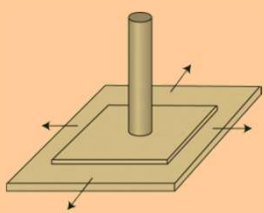
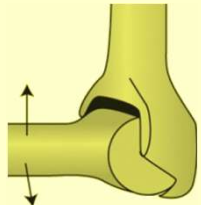
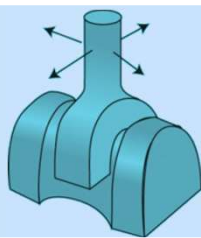
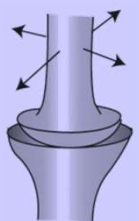
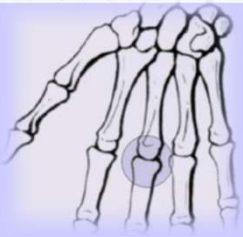
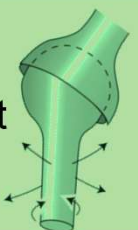


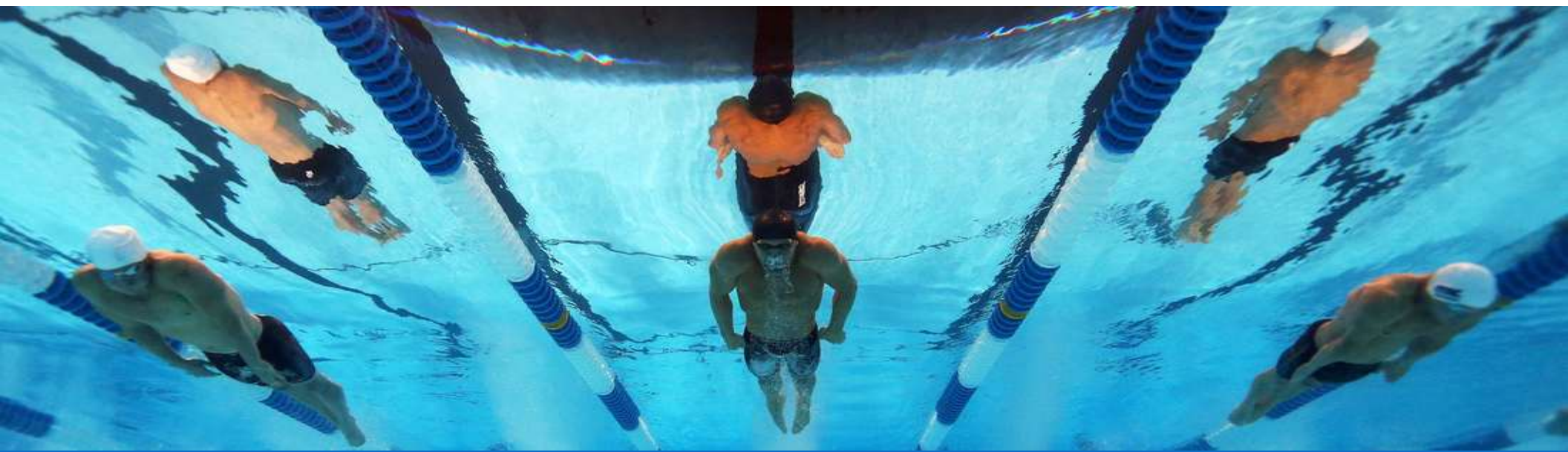
Multiaxial joints

- Movement about all three perpendicular axes

TYPES OF SYNOVIAL JOINTS



Joint type	Description	Movement	Examples
Pivot 	One bone rotates around one axis	Uniaxial: One-axis rotation	Neck 
Gliding 	Bone surfaces involved are nearly flat	Uniaxial : Gliding	Acromio-clavicular 
Hinge 	Convex and concave articulating surfaces	Uniaxial Flexion-extension	Elbow 
Saddle 	Bones set together as in sitting on a horse	Biaxial: Flexion-extension, abduction-adduction	Thumb 
Condylloid 	Ovular convex shape and reciprocal concave surfaces	Biaxial: Flexion-extension, abduction-adduction	Knuckles 
Ball and Socket 	A rounded bone is fitted into a cup-like receptacle	Multiaxial 3-axis rotation	Hip 



PUTTING IT ALL TOGETHER

- The interactions of the bones, muscles, and joints in the body allow motion to occur
- Healthy bones protect the body's organs and provide the framework for muscle attachment
- Muscles exert force to move the bones at the joints, resulting in the wide array of possible movements